Tutorial:
MDANSE basics

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MDANSE Workshop 2026

03/11/2026

Science and Technology Facilities Council
ISIS Neutron and Muon Source

Schedule: introduction

1. The origins of MDANSE.
2. The landscape of molecular dynamics analysis tools.
3. Design decisions: why do we “convert” a trajectory?
4. The relationship between MDANSE and MDANSE_GUI.

Schedule: practicals

1. Inspecting trajectory contents using MDANSE_GUI.
2. Running general-purpose analysis (PDF, DOS, etc.)
3. Saving and re-using scripts.
4. Isotropic and anisotropic systems.
5. Equilibration and simulation stability.

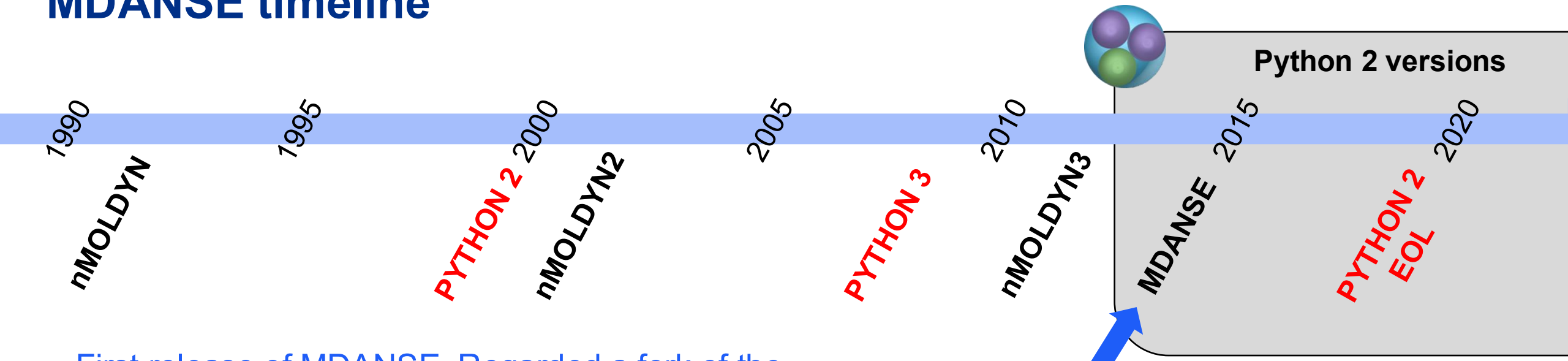
Software check

Please check that you have Python, any version from 3.11 to 3.14, and a virtual environment with:

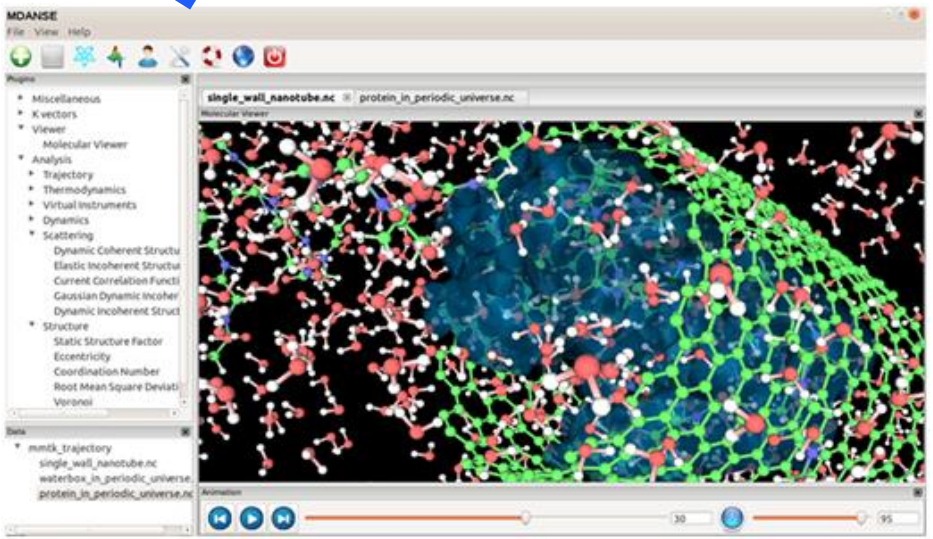
- mdanse[all]
- mdanse_gui[all]
- ase
- asap3

If any of those are missing, start installing them now

MDANSE timeline



- First release of MDANSE. Regarded a fork of the nMOLDYN3 program and incorporated code from ScientificPython and MMTK.*
- Not all features were retained in the transition from nMOLDYN to MDANSE.
- Originally developed at the ILL by E. Pellegrini et al.



JOURNAL OF CHEMICAL INFORMATION AND MODELING — Application Note — pubs.acs.org/jcim

MDANSE: An Interactive Analysis Environment for Molecular Dynamics Simulations

G. Goret, B. Aoun, and E. Pellegrini*

Computing for Science Group, Institut Laue-Langevin, 71 avenue des Martyrs, 38042 Grenoble, France

*https://github.com/ISISNeutronMuon/MDANSE/pull/169

MDANSE timeline

Python 2 versions

2021

2022

MDANSE 1.5.0

MDANSE 1.5.1

2023

MDANSE 1.5.2

2024

Python 3 versions

2025

MDANSE 2.0.0a1

MDANSE 2.0.0a2

MDANSE 2.0.0b1

2026

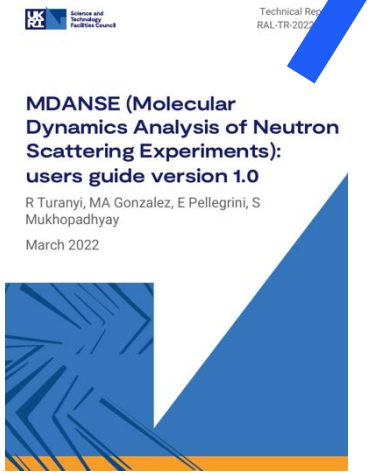
MDANSE 2.0.0

MDANSE 2.0.1

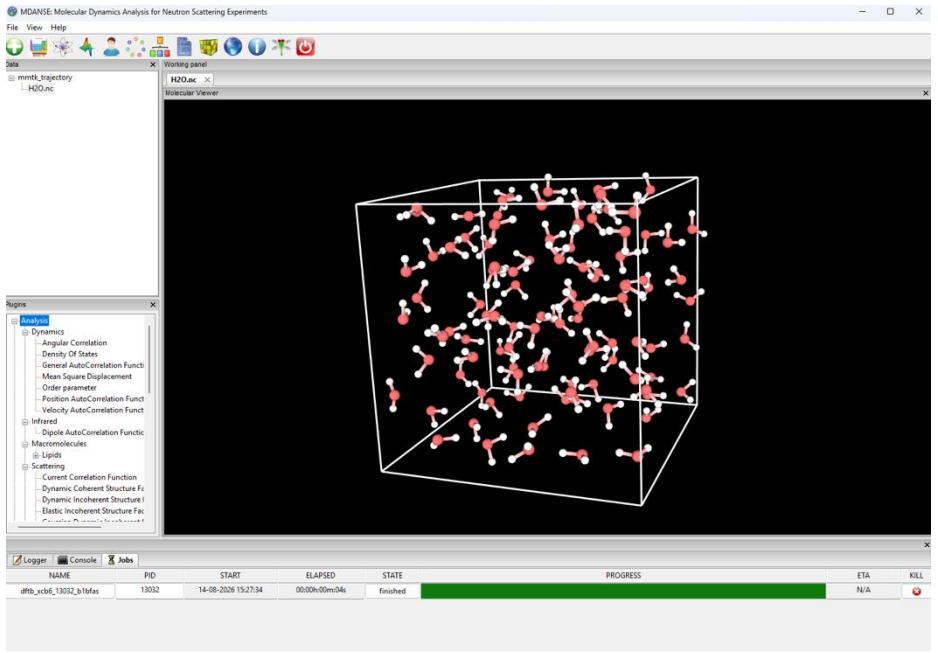
MDANSE 2.0.2

2027

MDANSE 2.x.y?



- A collaboration between software engineers and scientist at the ILL and ISIS to update MDANSE so that it can be installed and run on modern hardware.
- MDANSE repository moved to GitHub.
- MDANSE 1.5.2 was the last release which used Python 2.



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MDANSE timeline

Python 2 versions

2021

2022 — MDANSE 1.5.0, MDANSE 1.5.1

2023 — MDANSE 1.5.2

2024

Python 3 versions

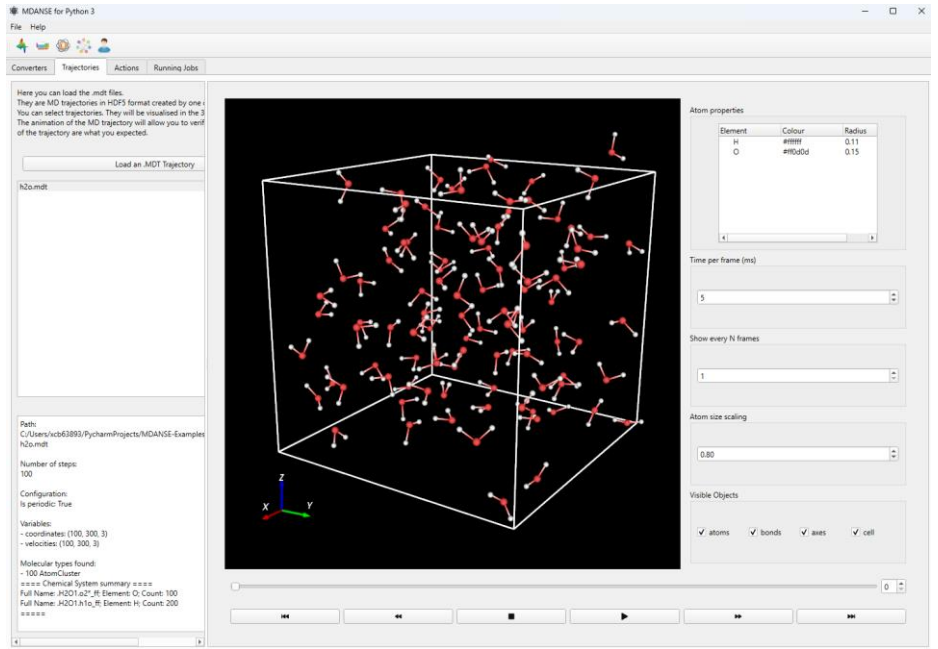
2025 — MDANSE 2.0.0a1, MDANSE 2.0.0a2

MDANSE 2.0.0b1

2026 — MDANSE 2.0.0, MDANSE 2.0.1, MDANSE 2.0.2

2027 — MDANSE 2.x.y?

- First alpha release of MDANSE 2 and the first release to use Python 3 which also included a complete overhaul of the front end to use Qt and VTK.
- Second alpha release included new plotting tools.



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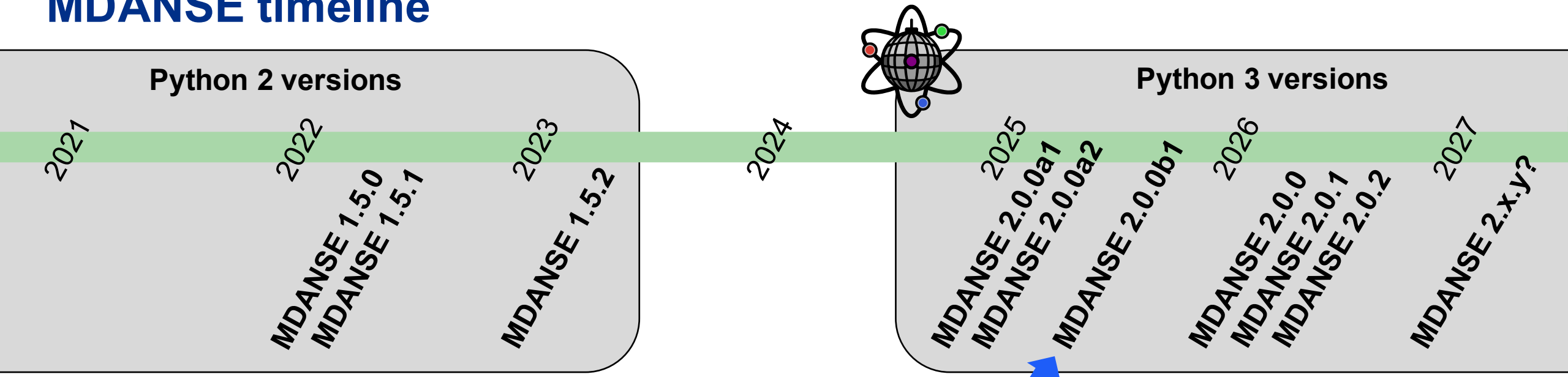
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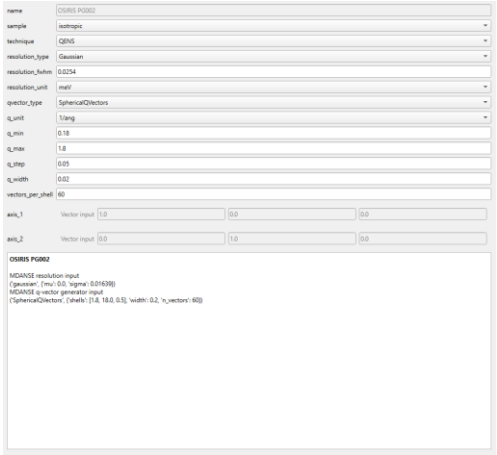
@isisneutronmuon

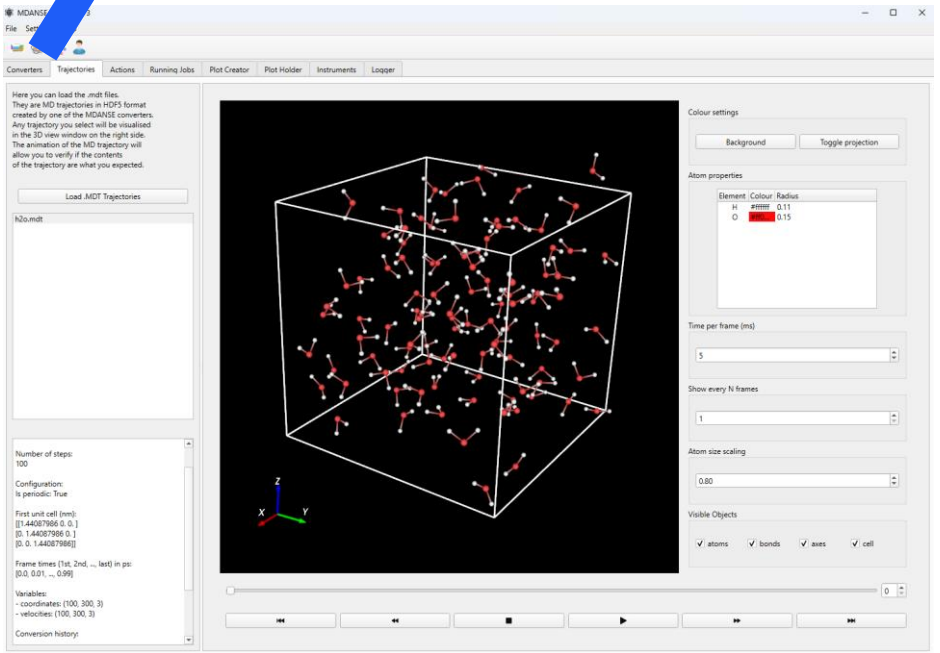
uk.linkedin.com/showcase/isis-neutron-and-muon-source

MDANSE timeline



- First beta release of MDANSE which included further fixes and improvements to the converter and analysis code.
- New analysis calculations (van Hove and current-correlation functions) and a new instrument profiles settings were added.





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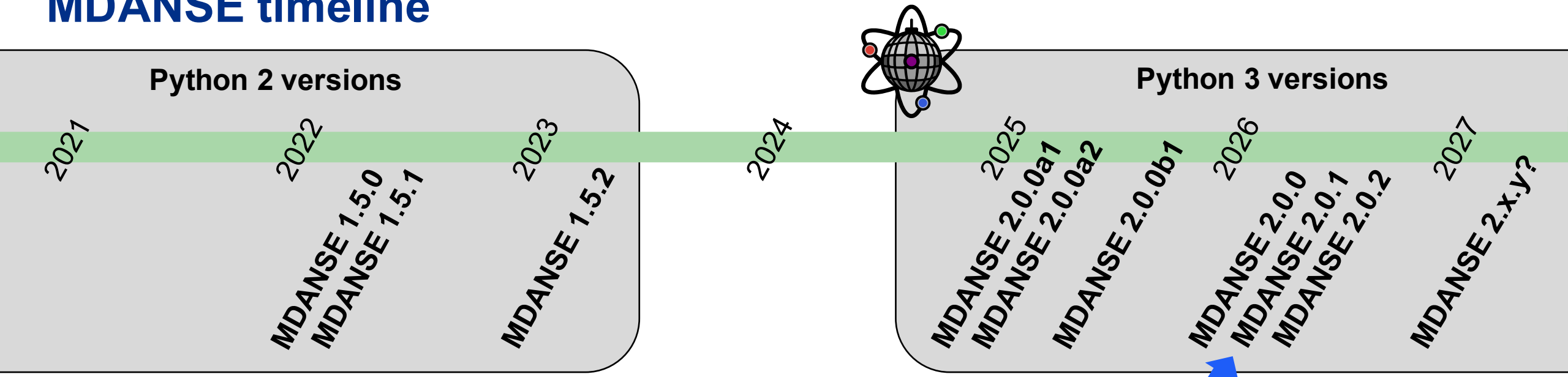
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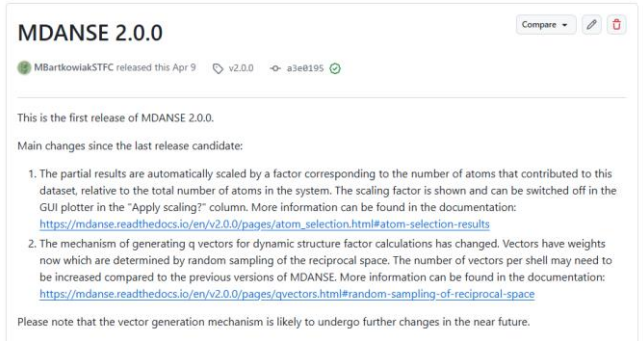
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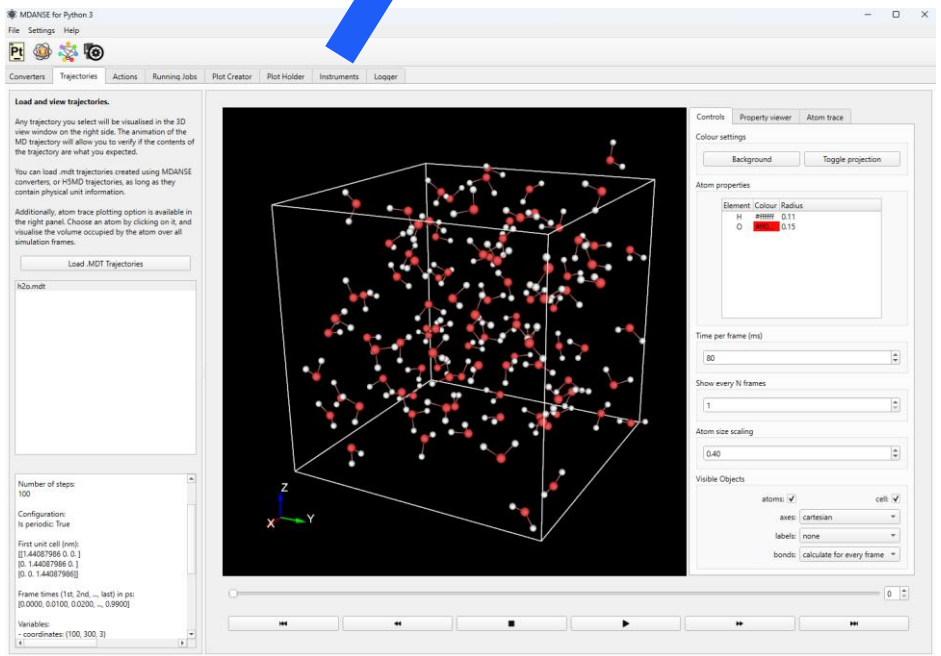
MDANSE timeline



- First official MDANSE 2 release which included a huge number of fixes, improvements and new features over the beta version.
- 2.0.1 included some further updates and fixes.
- Manuscript in preparation.







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Data check

In this tutorial we will need MD trajectories.

The plan is to generate them by running a script.

Start running the scripts called

`run_isotropic_trajectory.py`

`run_anisotropic_trajectory.py`

Molecular dynamics analysis tools – some alternatives

- There are many options to choose from.
- Pick your tools to match the problem you are solving.

Molecular dynamics analysis tools – some alternatives

- There are a lot of tools that can be used to view and analyze your molecular dynamics trajectories. Some of these may be better or worse for the research problem that you are trying to solve.

Neutron-related

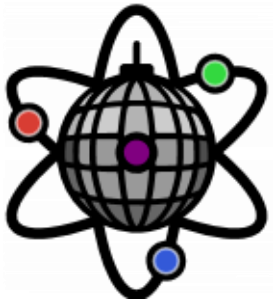




LiquidLib



dynasor



MDANSE

… OCLIMAX?

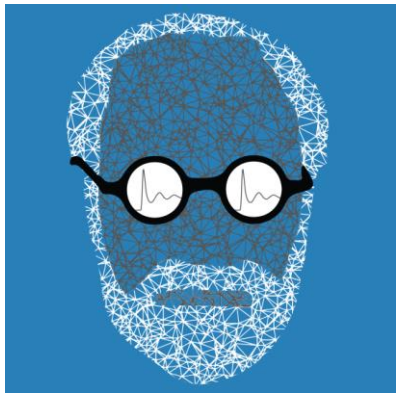
Molecular dynamics analysis tools – some alternatives

- There are a lot of tools that can be used to view and analyze your molecular dynamics trajectories. Some of these may be better or worse for the research problem that you are trying to solve.

General-purpose

MD ANALYSIS

MDTRAJ



Freud

mdapy

TRAVIS

VMD Visual Molecular Dynamics

MDANSE

Molecular dynamics analysis tools – some alternatives

- There are a lot of tools that can be used to view and analyze your molecular dynamics trajectories. Some of these may be better or worse for the research problem that you are trying to solve.

For GROMACS

CHAPERONg



StreaMD

Molecular dynamics analysis tools – some alternatives

- There are a lot of tools that can be used to view and analyze your molecular dynamics trajectories. Some of these may be better or worse for the research problem that you are trying to solve.

Has a GUI



OVITO



Moltwister



Optimised for speed…?







MDANSE basic workflow

Trajectory conversion

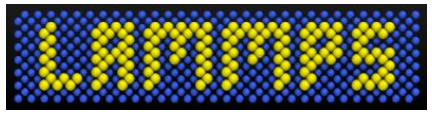












- Run a molecular dynamics simulation dump positions, velocity, etc.
- Convert trajectory to MDT (HDF5) format.

MDANSE basic workflow

Trajectory conversion

- Run a molecular dynamics simulation dump positions, velocity, etc.
- Convert trajectory to MDT (HDF5) format.

MD analysis



```
                self._nFrames,
            )
            self._outputData[f"j(q,t)_trans_{pair_str}"][index, :] = np.zeros(
                self._nFrames,
            )
        return

    rho_l, rho_t = x
    n_configs = self.configuration["frames"]["n_configs"]
    for pair_str, (label_i, label_j) in self.labels:
        corr_l = correlate(
            rho_l[label_i], rho_l[label_j][:n_configs], mode="valid"
        )[
            :,
            0,
            0,
        ] / (3 * n_configs * rho_l[label_i].shape[2])
        self._outputData[f"j(q,t)_long_{pair_str}"][index, :] += corr_l.real
        corr_t = correlate(
            rho_t[label_i], rho_t[label_j][:n_configs], mode="valid"
        )[
            :,
            0,
            0,
        ] / (3 * n_configs * rho_t[label_i].shape[2])
        self._outputData[f"j(q,t)_trans_{pair_str}"][index, :] += corr_t.real

def finalize(self):
    """Normalize, Fourier transform and write the results out."""
```

- Load trajectory and run analysis calculation.
- Save results as MDA (HDF5) or other formats.

MDANSE basic workflow

Trajectory conversion

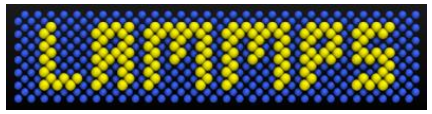












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MD analysis

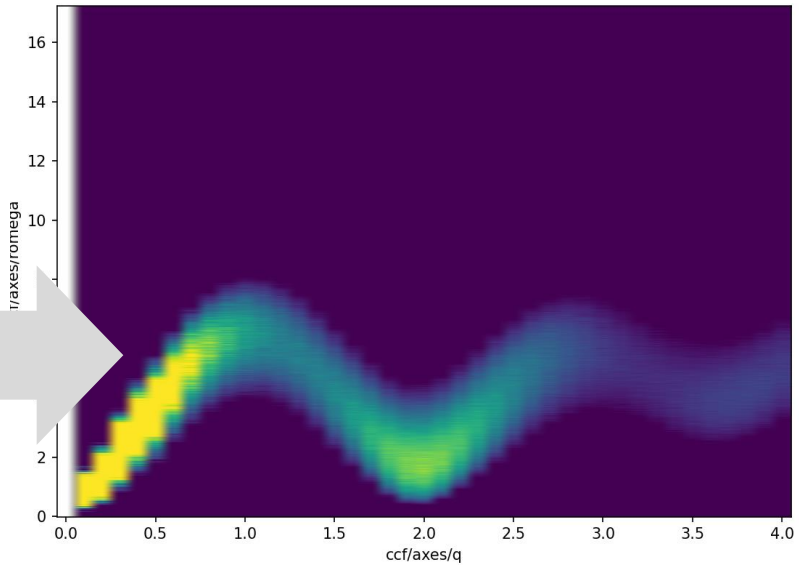
```
                self._nFrames,
            )
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    rho_l, rho_t = x
    n_configs = self.configuration["frames"]["n_configs"]
    for pair_str, (label_i, label_j) in self.labels:
        corr_l = correlate(
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        )[
            :,
            0,
            0,
        ] / (3 * n_configs * rho_l[label_i].shape[2])
        self._outputData[f"j(q,t)_long_{pair_str}"][index, :] += corr_l.real
        corr_t = correlate(
            rho_t[label_i], rho_t[label_j][:n_configs], mode="valid"
        )[
            :,
            0,
            0,
        ] / (3 * n_configs * rho_t[label_i].shape[2])
        self._outputData[f"j(q,t)_trans_{pair_str}"][index, :] += corr_t.real

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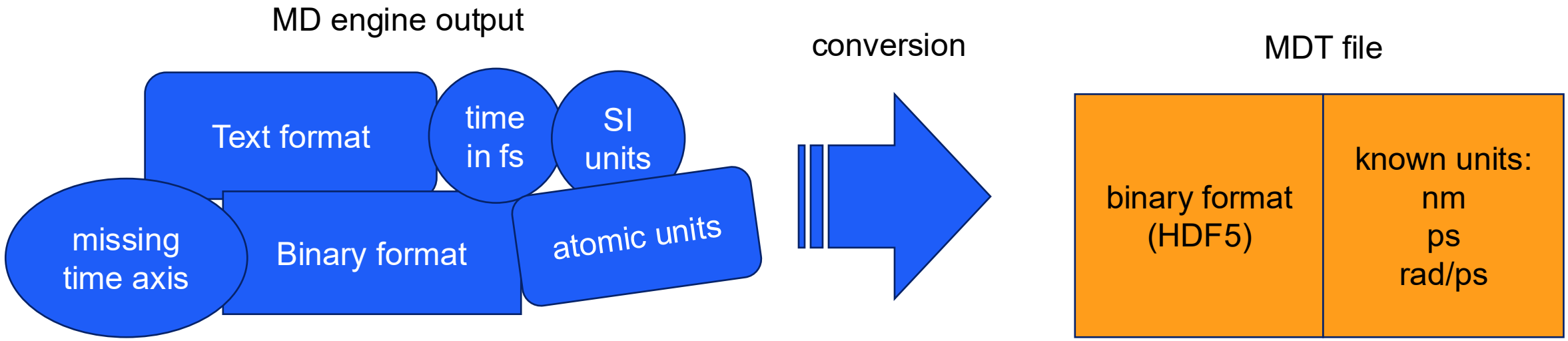
Plotting



- Load (MDA format) analysis results and plot using MDANSE GUI or with other tools/libraries.

MDANSE converters

- A few general converters have been included, which use other MD analysis libraries to convert them to the MDANSE format (HDF).
- This should mean most MD topology and trajectory files can now be converted.



Trick question time

- We just used ASE to run MD simulations and write out trajectories. Which converter should we use to get the trajectories into MDANSE?
- The MDANSE GUI is divided into tabs. If you're looking for converters, which is the right tab to go to?



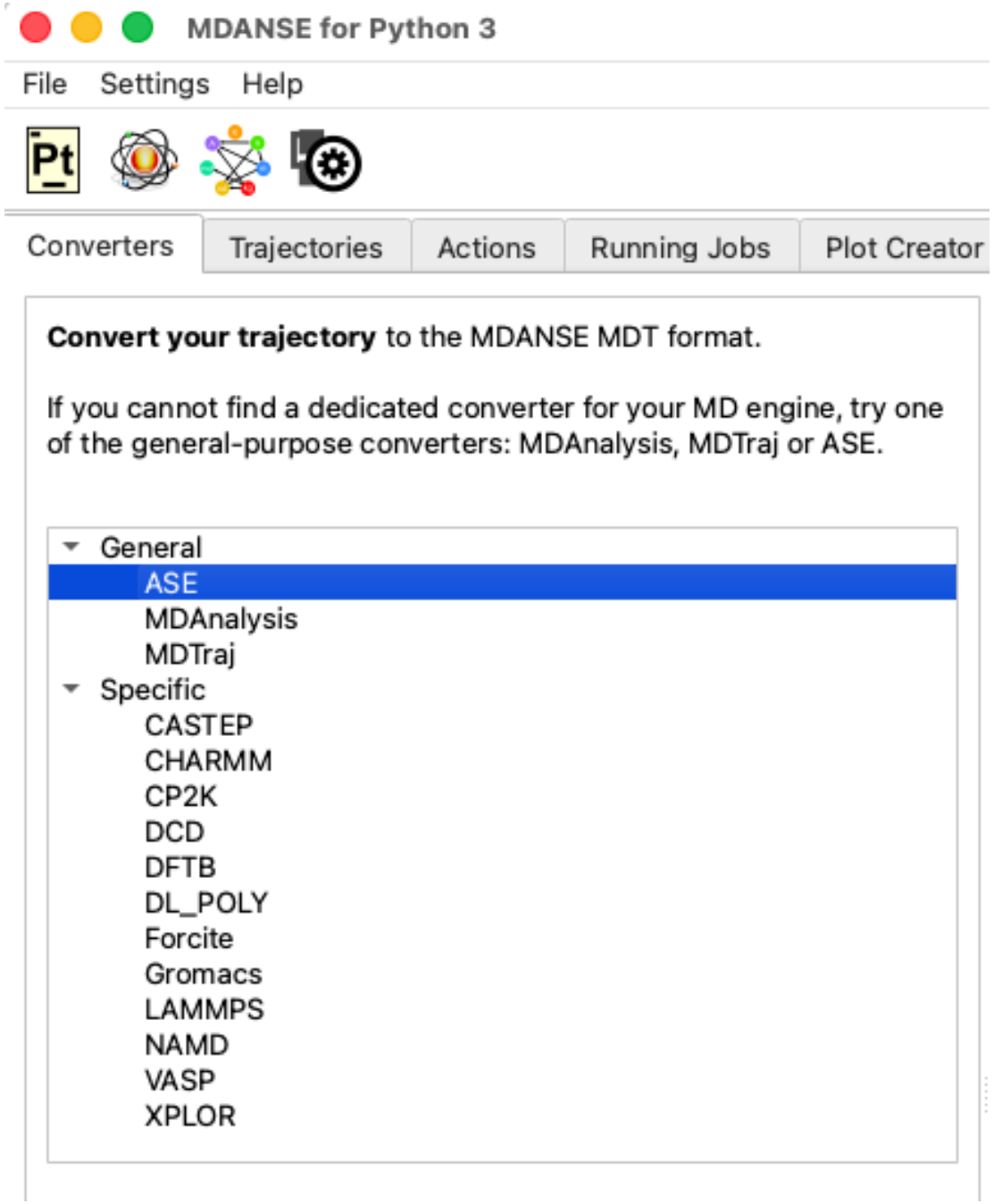
Trick question time

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ASE converter

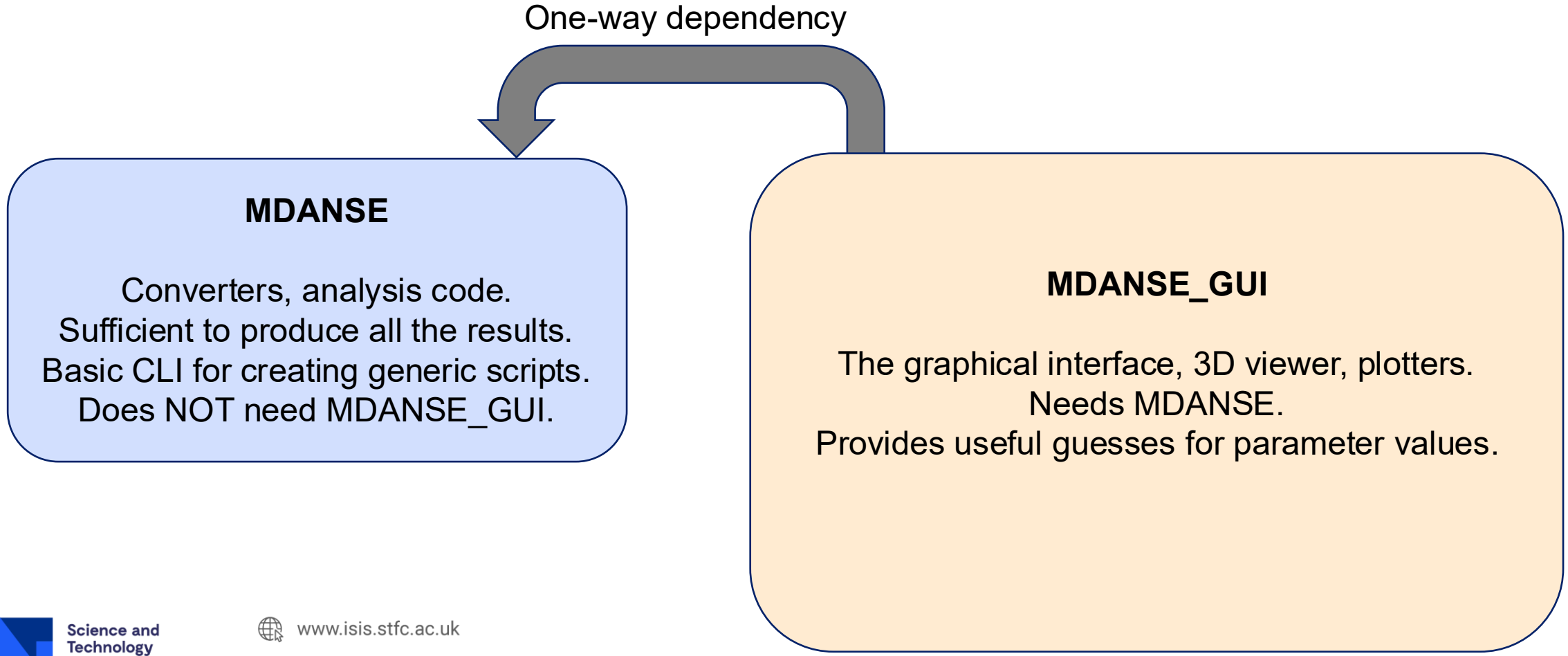
- The MDANSE GUI is divided into tabs. If you're looking for converters, **which is the right tab** to go to?

Converters tab, but we also accept the answer "the first one".



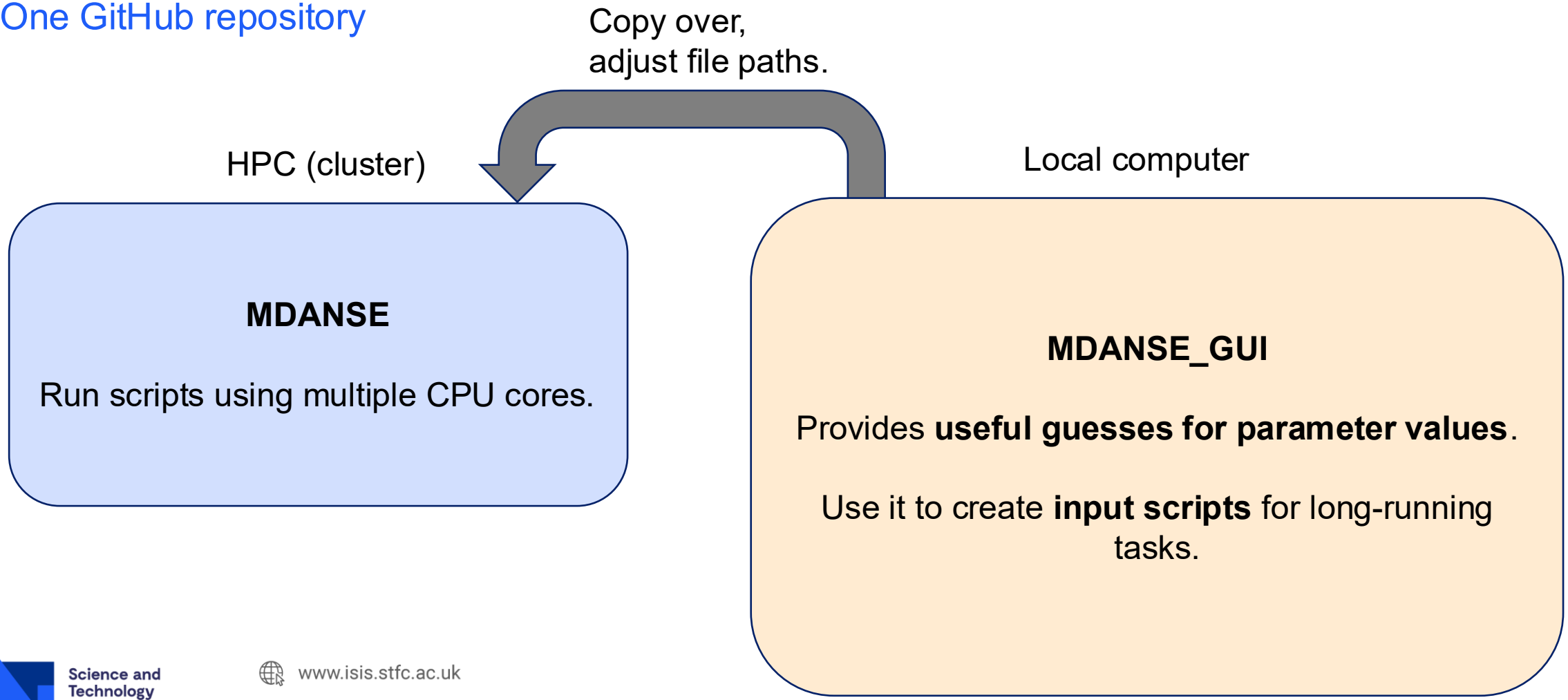
MDANSE and MDANSE_GUI

- Two Python packages
- One GitHub repository



MDANSE and MDANSE_GUI: a use case.

- Two Python packages
- One GitHub repository



MDANSE Basics: practicals.

We're about to start the GUI.

You're welcome to run

`mdanse_gui -h`

to find out that some things can be customised.

You can also run

`create_mdanse_link`

if you like having a desktop icon for the GUI.

You can just start the GUI by running

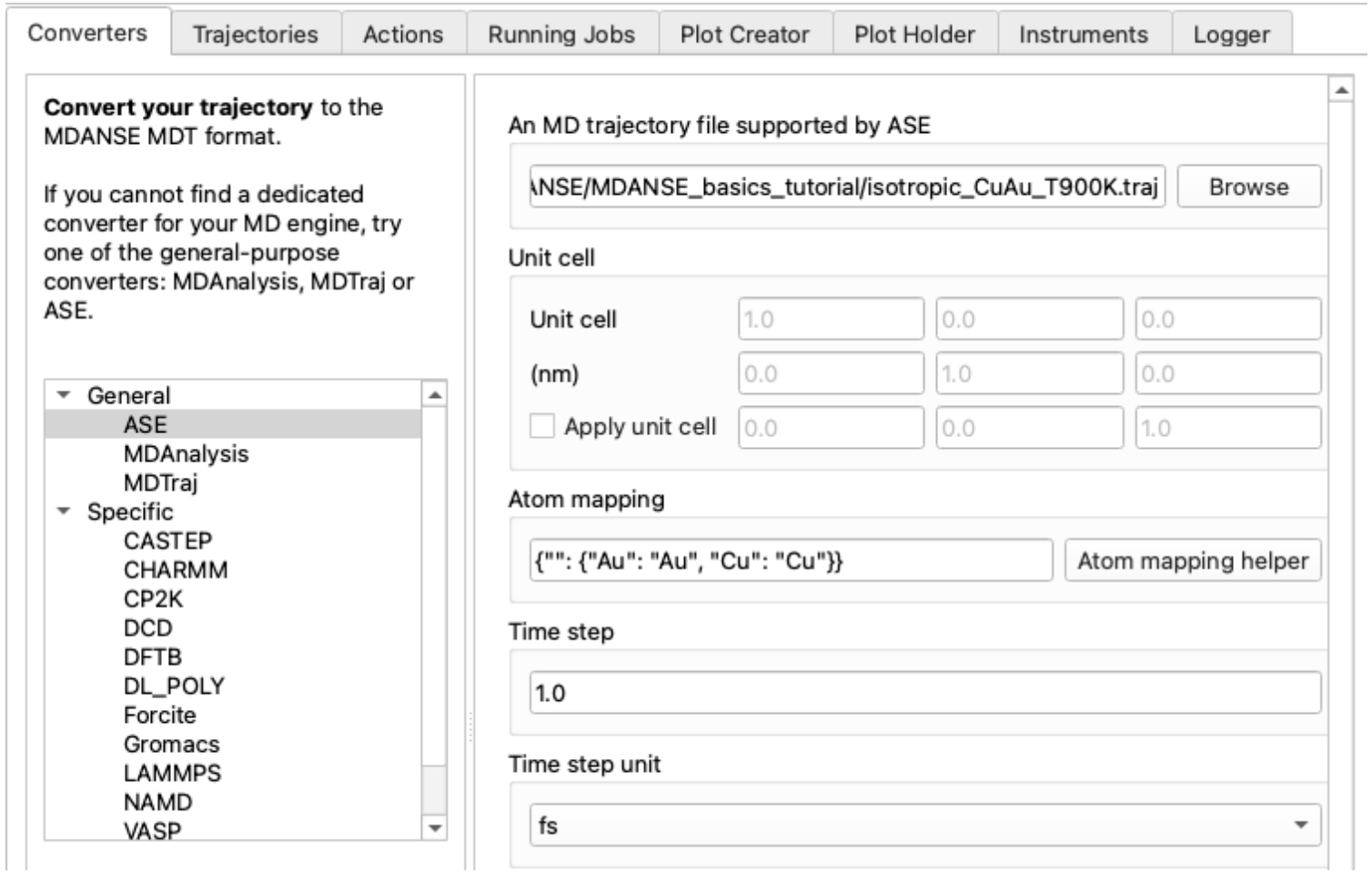
`mdanse_gui`

MDANSE Basics: practicals.

1. Convert both trajectories we just created.

- Check the input parameters. Some of them may need to be set manually to correct values.
- If you're not sure about some values, check the original script used for running the MD simulation.

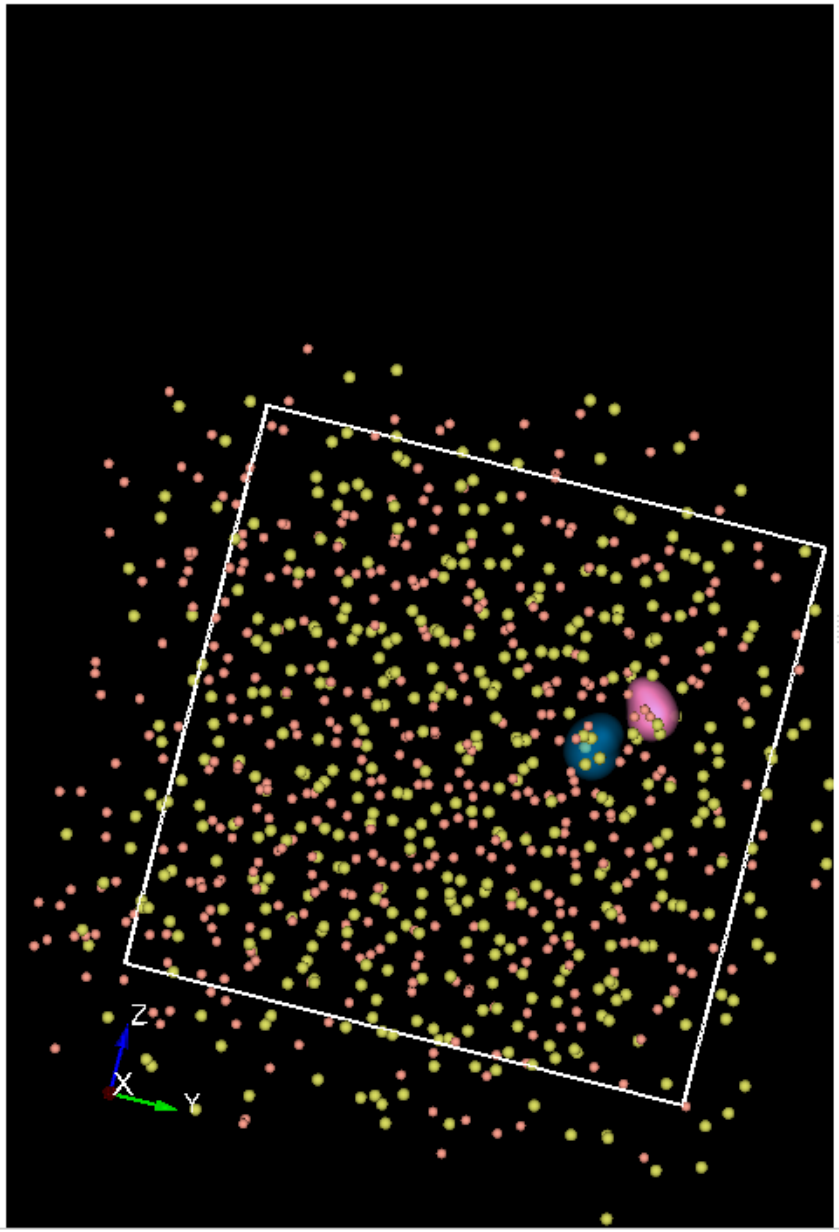
```
timestep=5.0 * units.fs,
```

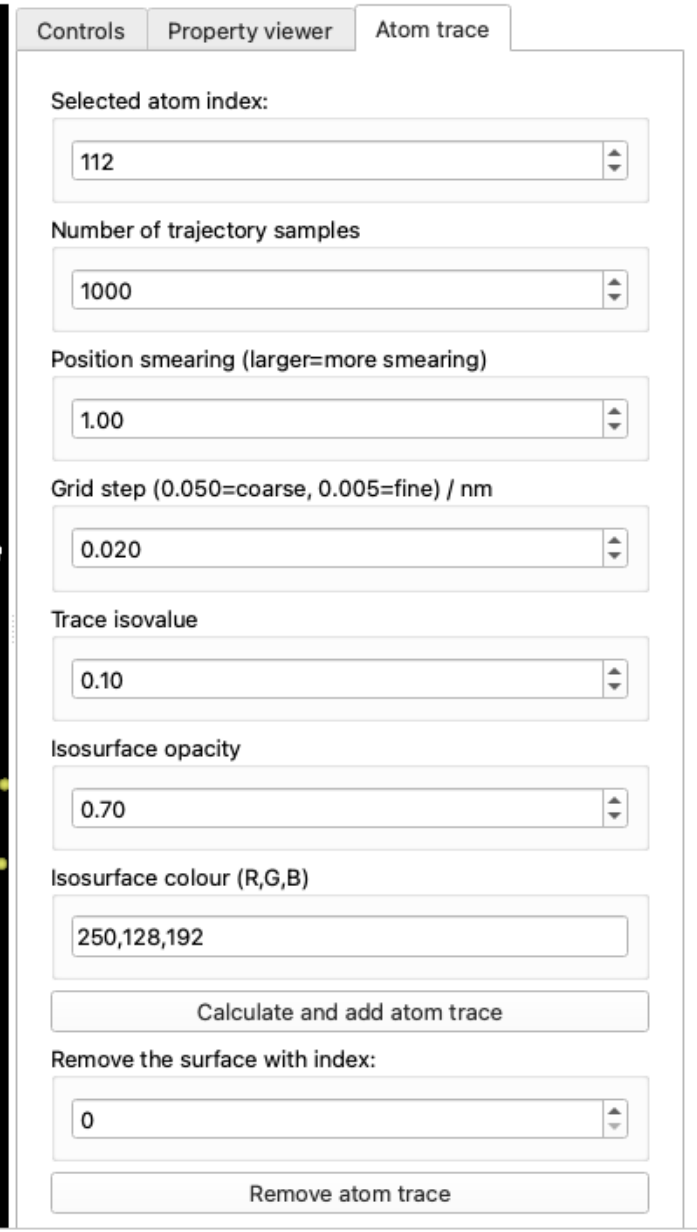


MDANSE Basics: Trajectory tab

Have a look at your trajectory:

- Switch off the bonds in the 3D view.
- Watch the animation of your trajectory.
- See how far an atom has moved: calculate the atom trace for some atoms.





MDANSE Basics: Trajectory tab

Find some numbers:

- How many atoms are in the trajectory?
- What are the first and last values on the time axis?
- What are the dimensions of the unit cell?
- What are the electric charge values on the atoms?

MDANSE Basics: GUI history

1. Switch off the GUI.
2. Start it again.
3. Go to File -> Open Recent Trajectories
4. Bring back the trajectories you were using before.

MDANSE Basics: GUI settings

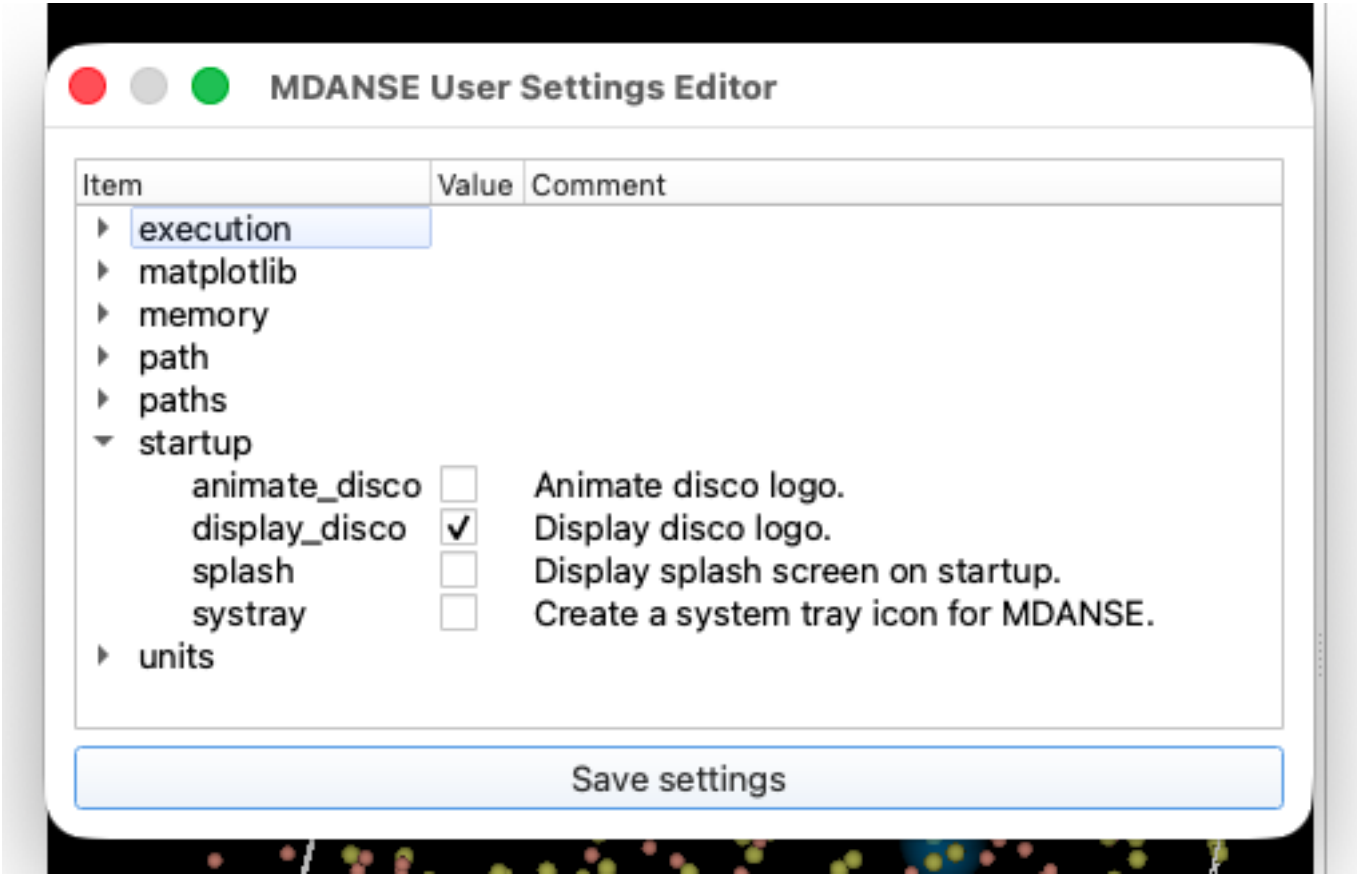
Was it difficult to close the GUI?

Were you not happy with the splash screen?

Did you find the 3D placeholder distracting?

Don’t just tolerate it!

Do something about it!



MDANSE Basics: plotting

Calculate and plot the temperature for both trajectories

MDANSE Basics: plotting

Calculate and plot the temperature for both trajectories

Once the calculation is done, try creating a quick plot by double-clicking the file name in Plot Creator.

Bonus task:

Create a new, empty plot and use it to show both temperature curves in a single figure.

MDANSE Basics: scripting

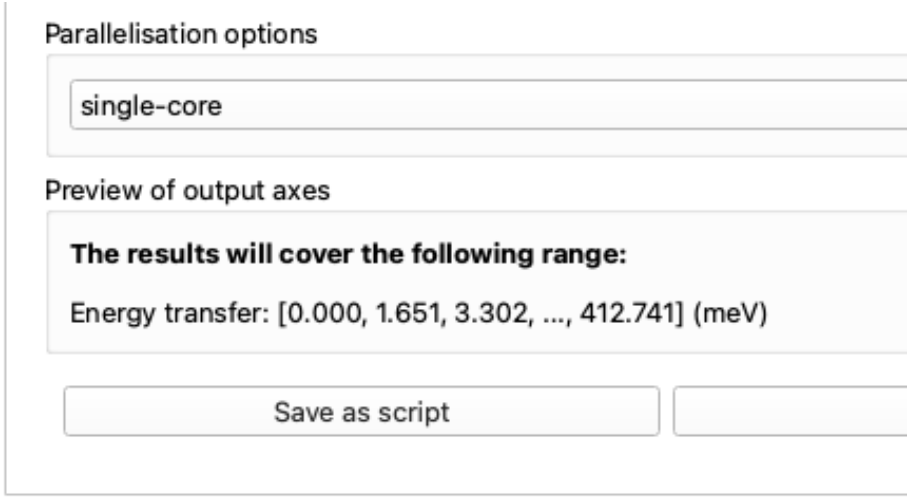
Calculate the vibrational density of states (vDOS) of both trajectories, using axial projection.

We want to calculate the energy spectra of vibrations along specific directions in the system.

We will use projection vectors 100, 010 and 001.

The aim of this task is to use **a script** to perform the calculations.

MDANSE Basics: scripting



We will use projection vectors 100, 010 and 001.

The aim of this task is to use **a script** to perform the calculations.

Steps:

1. In the GUI, go to the Actions tab and select Dynamics->DensityOfStates.
2. Change the projection input to AxialProjection and set the vector to 100.
3. Instead of running the analysis, click “Save as script”. Save the Python script with a meaningful name.
4. Edit the script. Make the script run the analysis 6 times, using 3 different input vectors and 2 different input trajectories.

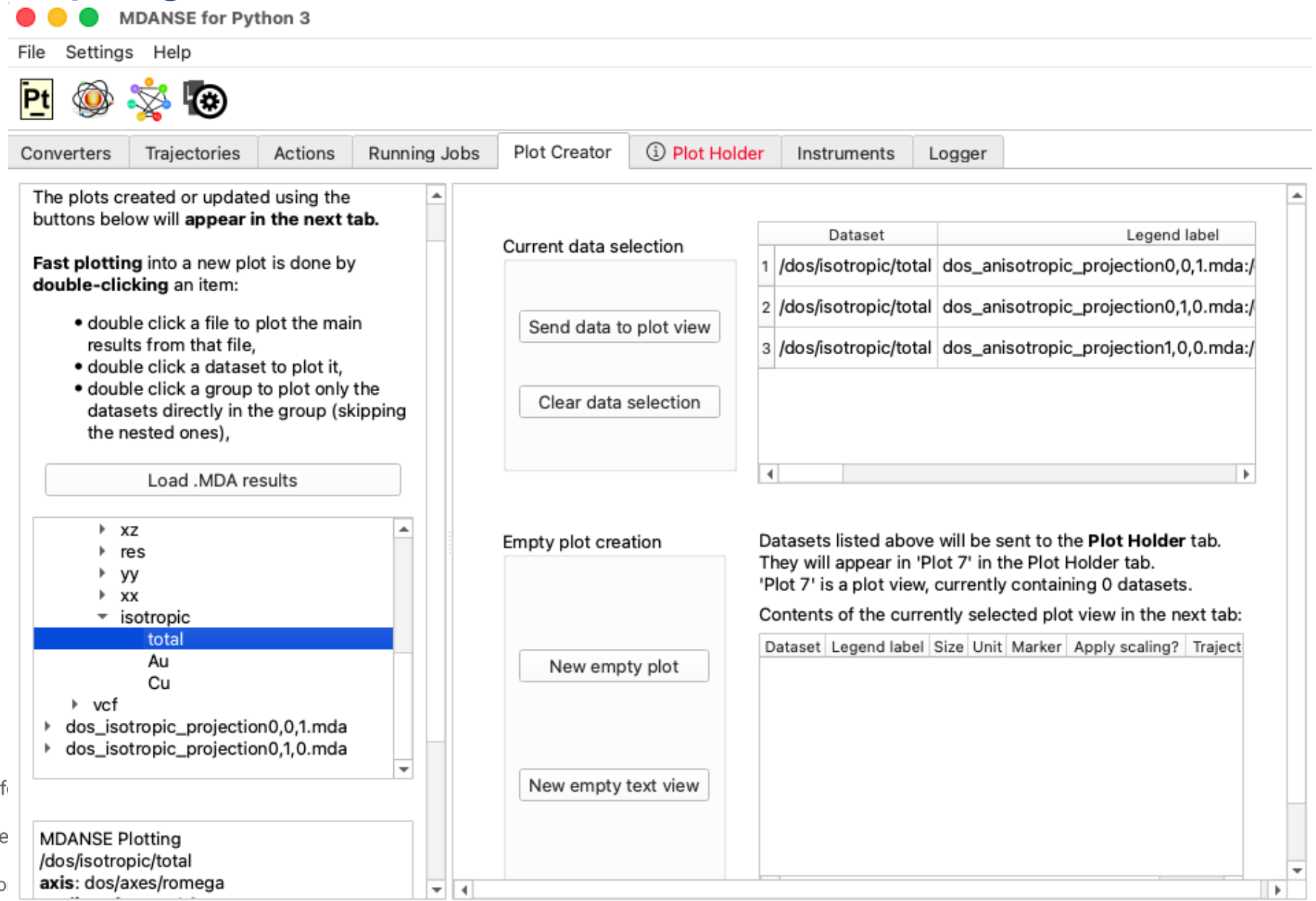
Make sure your script changes the output file name between the analysis runs!

MDANSE Basics: scripting

Run the script. It should create 6 output files.

Load all of them into the GUI in the Plot Creator tab.

Select the total isotropic DOS from different files and send the results to an empty plot.

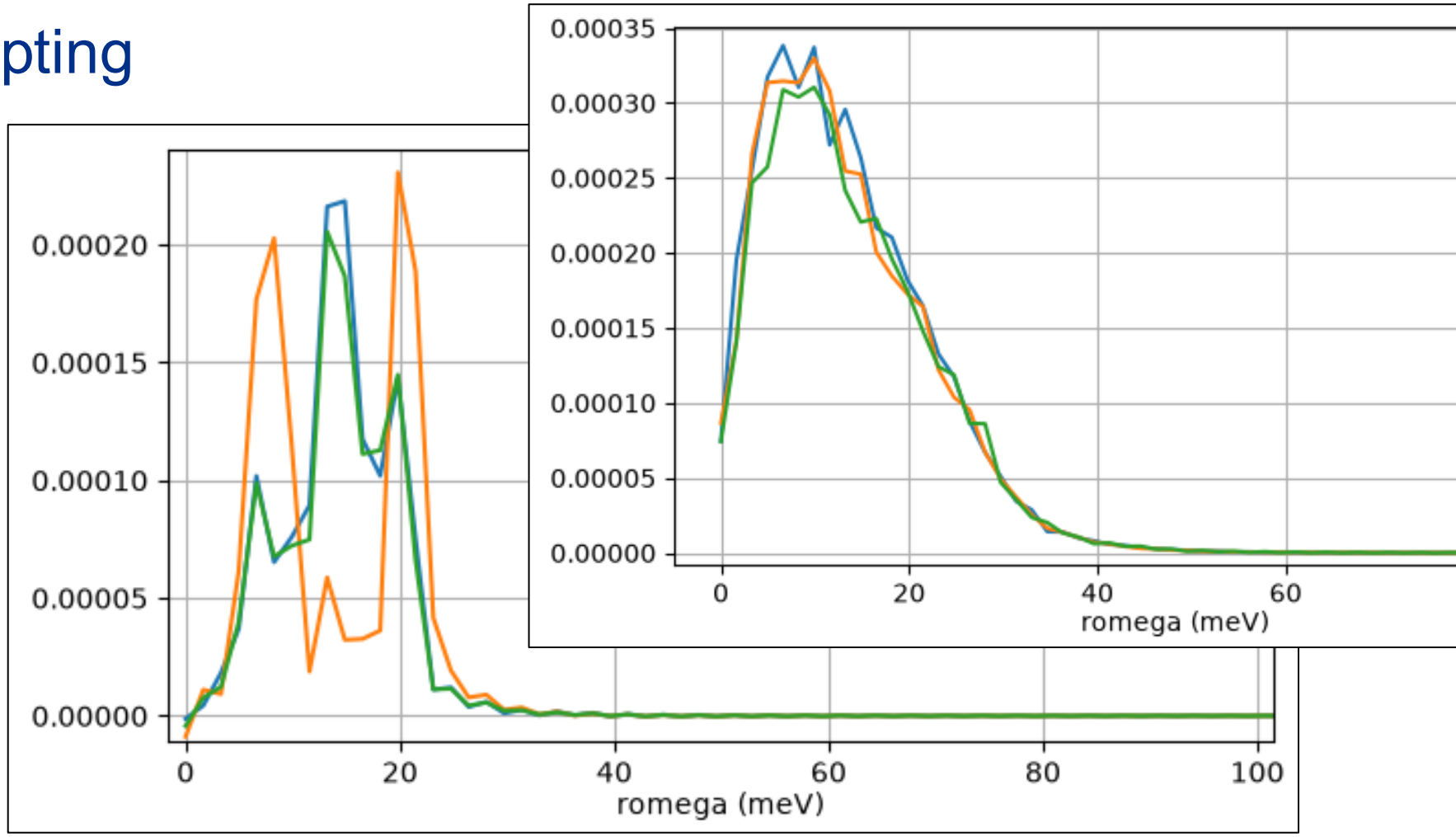


MDANSE Basics: scripting

Run the script. It should create 6 output files.

Load all of them into the GUI in the Plot Creator tab.

For each trajectory, select the total isotropic DOS from different files and send the results to an empty plot.



Did you get plots similar to this one?

Can you explain the difference between the curves?

MDANSE Basics: more plotting

For each trajectory, run the StaticStructureFactor3D analysis.

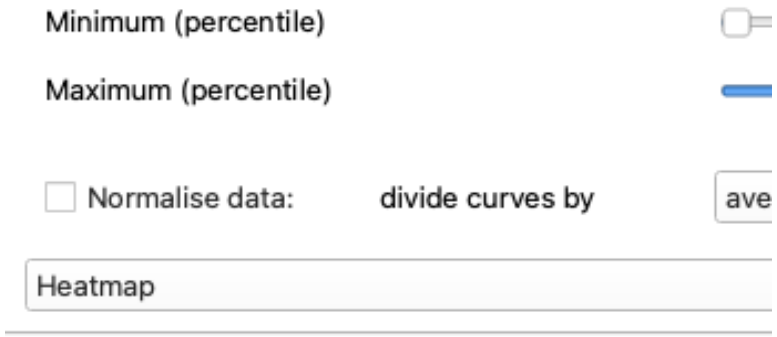
The most important part is the definition of the q-vector grid.

3D Q vector generator

1	seed	0	IntegerConfig...
2	n_samples	100	IntegerConfig...
3	supercell_used	(6, 6, 6)	VectorConfig...
4	q1	(1.0, 0.0, 0.0)	QVectors3DV...
5	q2	(0.0, 1.0, 0.0)	QVectors3DV...
6	q3	q1 x q2	QVectors3DV...
7	q1_range	(-3.0, 3.0, 0.166)	RangeConfig...
8	q2_range	(-3.0, 3.0, ...	RangeConfig...
9	q3_range	(0.0, 0.0, 0.1)	RangeConfig...

MDANSE Basics: more plotting

Plot the results of the SSF3D calculation as Heatmap.



You will have to change the main axis of the dataset, and the index range used for plotting.

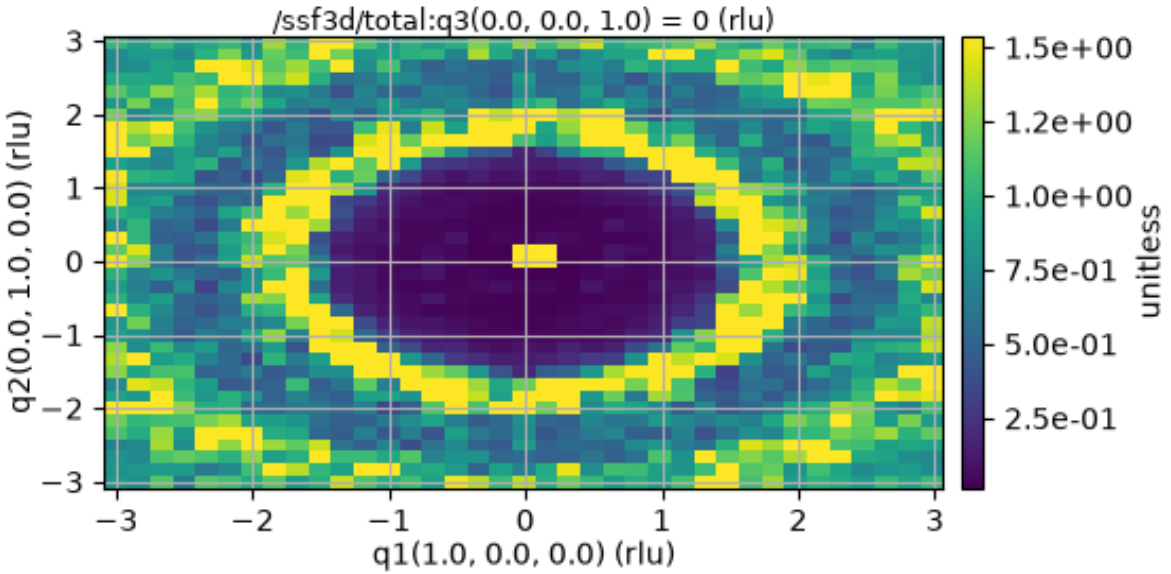
	Legend label	Size	Unit	Main axis	Use it?
3d/total	diffraction_is...	(37, 37, 1)	unitless	q3(0.0, 0.0, 1....	✓ 0
3d/total	diffraction_an...	(37, 37, 1)	unitless	q3(0.0, 0.0, 1....	✓ 0

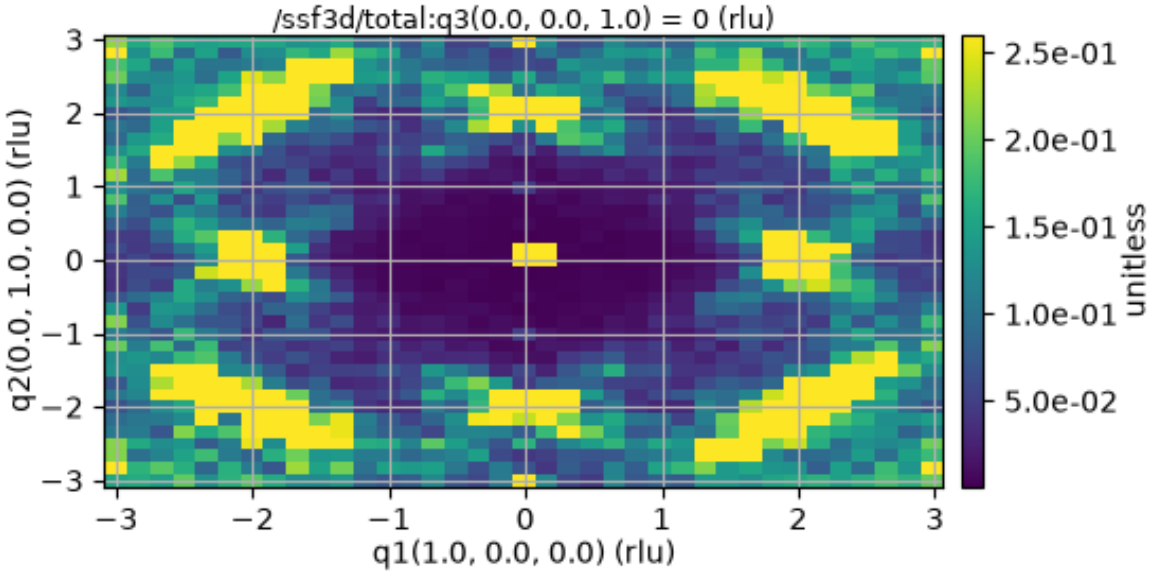
MDANSE Basics: more plotting

Adjust the sliders under the plot.

It will change the limits of the colour map.

Did you get a result like this?





This concludes the main part of the tutorial.

The tutorial on trajectory stability can follow here, depending on the remaining time.